

## Solution Requirements

|                      |                                       |
|----------------------|---------------------------------------|
| <b>Date</b>          | 27 June 2026                          |
| <b>Team ID</b>       | XXXXXX                                |
| <b>Project Name</b>  | A Comprehensive Measure of Well-Being |
| <b>Maximum Marks</b> | 4 Marks                               |

### Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

### Step 1: Functional Requirements (FR)

| S.No | Requirement Category | Requirement Description                                                                                                  | Priority (High/Medium/Low) |
|------|----------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 1    | Authentication       | Users shall securely log in to the system using username and password credentials.                                       | High                       |
| 2    | Authorization levels | Different users such as Researchers, Policymakers, and Administrators shall have role-based access permissions.          | Medium                     |
| 3    | External interfaces  | The system shall interact with external datasets and machine learning libraries such as Pandas, Scikit-learn, and Flask. | Medium                     |

|   |                                   |                                                                                                                |        |
|---|-----------------------------------|----------------------------------------------------------------------------------------------------------------|--------|
| 4 | Transactions processing           | The system shall accept HDI indicator values, process them through the ML model, and generate predictions.     | High   |
| 5 | Reporting                         | The system shall generate HDI prediction reports, visualizations, and downloadable analysis results.           | High   |
| 6 | Business rules, etc.              | The system shall predict HDI scores based on life expectancy, education indicators, and GNI per capita values. | High   |
| 7 | Compliance to laws or regulations | The system shall ensure secure handling of user data and comply with applicable data privacy guidelines.       | Medium |
| 8 | <i>Other</i>                      | The system shall allow users to compare HDI predictions across different countries.                            | Medium |

**Step 2: Non-Functional Requirements (NFR)**

| S.No | NFR Category        | Requirement Description                            | Target Metric / Acceptance Criteria                 |
|------|---------------------|----------------------------------------------------|-----------------------------------------------------|
| 1    | Performance & Speed | The system shall generate HDI predictions quickly. | Prediction results displayed in less than 3 seconds |

|          |                            |                                                                      |                                                   |
|----------|----------------------------|----------------------------------------------------------------------|---------------------------------------------------|
| <b>2</b> | Scalability                | The system shall support increasing numbers of users and datasets.   | Supports at least 100 concurrent users            |
| <b>3</b> | Security & Data Privacy    | The system shall protect user information and prediction data.       | Secure authentication and encrypted data storage  |
| <b>4</b> | Reliability & Availability | The system shall be available for users with minimal downtime.       | 99% uptime per month                              |
| <b>5</b> | Usability & Accessibility  | The system shall provide an easy-to-use web interface for all users. | User-friendly interface with intuitive navigation |
| <b>6</b> | <i>Other</i>               | The system shall be easy to update and maintain.                     | Modular code structure with proper documentation  |